
paper_id: PAPER_998
title: "REST F_U_Bi Gamma Sweep Endpoints — /api/fubi/inside-outside + /api/fubi/gamma-sweep"
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calibration: {port: 3141, routes: 18, new_routes: 2}
sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_998: REST F_U_Bi Gamma Sweep Endpoints

Abstract

We add two new REST API endpoints to `uqff_server.js` (Port 3141), bringing the total to 18 routes:

1. POST /api/fubi/inside-outside

Computes the F_U_Bi inside-to-outside buoyancy mass portion.

Request

```
{"M_kg": 1.989e30, "r_m": 1.496e11}
```

Response

```
{  
  "F_U_Bi": 2.33e40, "ratio": 0.606,  
  "Ug": 5.93e-03, "Ub": -3.57e-03,  
  "rho_Scm": 1.0e-10, "V_region": 1e48, "S26": 19.56  
}
```

2. POST /api/fubi/gamma-sweep

Computes the aggregate F_U_Bi across systems at multiple Γ values.

Request

```
{"gamma_THz_list": [0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0]}
```

Response

```
{  
  "systems": 20, "gamma_count": 7,  
  "results": [{"gamma_THz": 0.01, "aggregate": -6.11e13}, ...]  
}
```

3. Implementation

File: `uqff_server.js`, routes 17–18. CP4 class #582.